Bluetooth, infrared and always-on control bridge

Rev F integrated keyboard computer. See validation reports.

ESP32-C3 Bluetooth / serial bridge

File: bridge.kicad_sch

Keyboard, power, rotary and indicator harnesses

File: interfaces.kicad_sch

Programming and boot control

File: program.kicad_sch

Learn and transmit infrared

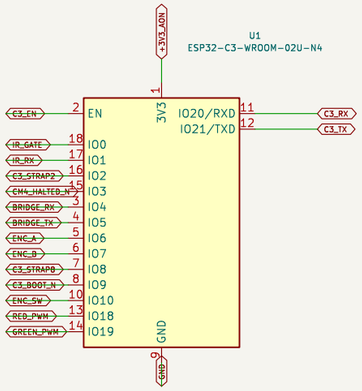
File: infrared.kicad_sch

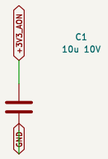
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File: control.kicad_sch

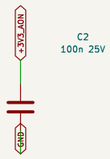
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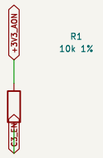
Size: A2	Date:	**Rev:**
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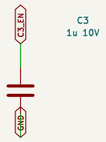
AON 3.3V: BLE only; Wi-Fi runs on CM4. GPIO2/8/9 boot straps retain pullups.
UART1 GPIO4 TX and GPIO5 RX connect to RP2040; UART0 GPIO20 RX and GPIO21 TX connect to CM4.
External 2.4GHz antenna must sit in plastic away from the key plate, battery and cables.

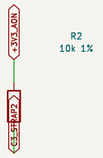


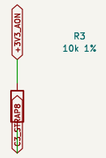


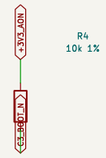












Connected schematic; hardware and manufacturing validation pending

Sheet: /ESP32-C3 Bluetooth / serial bridge/
File: bridge.kicad_sch

Title: ESP32-C3 Bluetooth / serial bridge

Size: A3	Date:	**Rev: F**
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J1/J2 pass the RP2040 I2C master bus through to the power board without a split cable.
UART names are referenced to RP2040: BRIDGE_RX goes to ESP32-C3 TX.
J3 CM4 UART/control signals are isolated on the CM4 carrier during power off.

J1
Keyboard J4



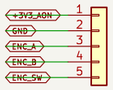
J2
Power J5



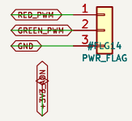
J3
CM4 carrier J7



J4
Rotary J1



J5
Battery light J1



J7
CM4 carrier J10 AON



Connected schematic; hardware and manufacturing validation pending

Sheet: /Keyboard, power, rotary and indicator harnesses/
File: interfaces.kicad_sch

Title: Keyboard, power, rotary and indicator harnesses

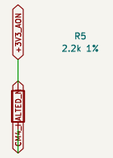
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KiCad E.D.A. 9.0.7 | Id: 3/5

UART0 programming header uses 3.3V logic. Never connect 5V serial signals.
Hold BOOT while pulsing RESET for UART ROM download; the CM4 can drive these open-drain lines.
2.2k pullup on HALTED_N: carrier NMOS pulls low only after the Linux kernel reaches final halt.









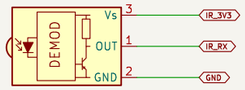
Connected schematic; hardware and manufacturing validation pending

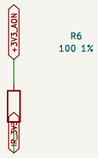
Sheet: /Programming and boot control/
File: program.kicad_sch

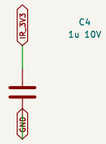
Title: Programming and boot control

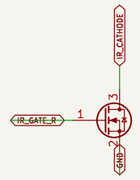
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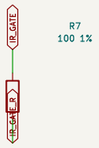
38kHz demodulating receiver records pulse timings from the ORIGINAL REMOTE.
Two 940nm emitters use independent 33ohm resistors; firmware caps duty and transmission length.
Learned / trial commands are saved only after the user confirms success; no passive device readout is possible.

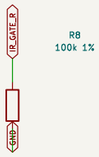




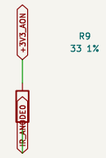






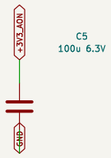






















Connected schematic; hardware and manufacturing validation pending

Sheet: /Learn and transmit infrared/
File: infrared.kicad_sch

Title: Learn and transmit infrared

Size: A3	Date:	**Rev: F**
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